

Phase 2 -- Error Analysis

Rotten Tomatoes test set

Model: fabriceyhc/bert-base-uncased-imdb
(no fine-tuning -- see EVALUATION_SUMMARY.md)

Test set size: 1,066 reviews
Accuracy: 0.8096

Error breakdown

Total errors: 203 (19.0%)
False positives (true negative -> predicted positive): 152
False negatives (true positive -> predicted negative): 51

Confidence (overconfidence check)

Correct predictions -- mean confidence 0.993
Errors (all) -- mean confidence 0.985
false positives -- mean confidence 0.985
false negatives -- mean confidence 0.985
Errors made with confidence ≥ 0.90 : 193/203 (95.1%)

The model is almost as confident when wrong as when right -- this matches the Phase 1 calibration FAIL (mean conf 0.993, near-certain on everything). Confidence is not a useful signal for filtering bad predictions here.

Review length (words)

Correct -- mean 21.2 median 20
Errors -- mean 21.5 median 21

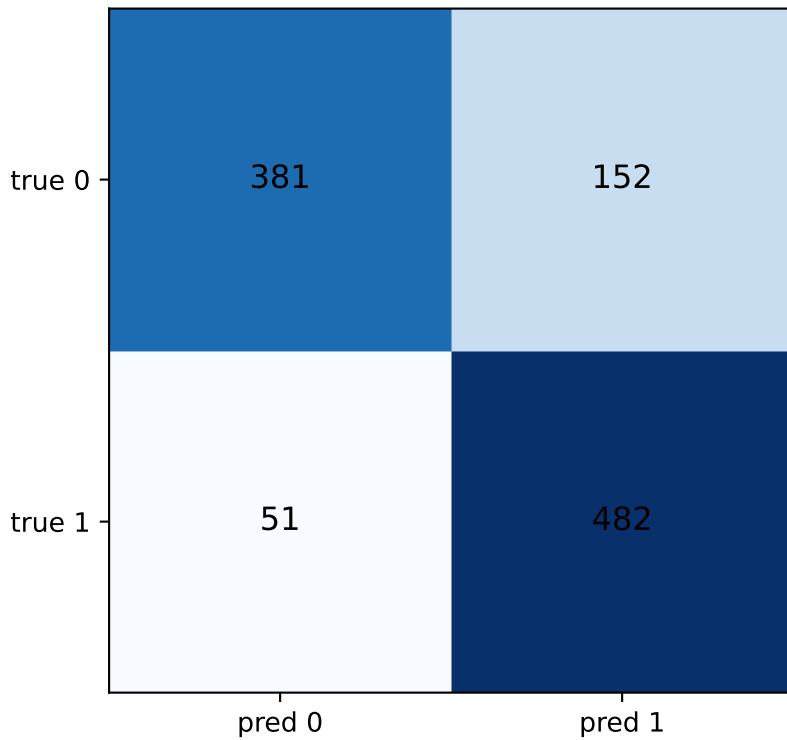
Lexical cues (share of reviews containing the cue)

Negation -- correct 15.8% | errors 12.8% | false positives 13.2%
Contrast -- correct 20.6% | errors 24.6% | false positives 25.0%

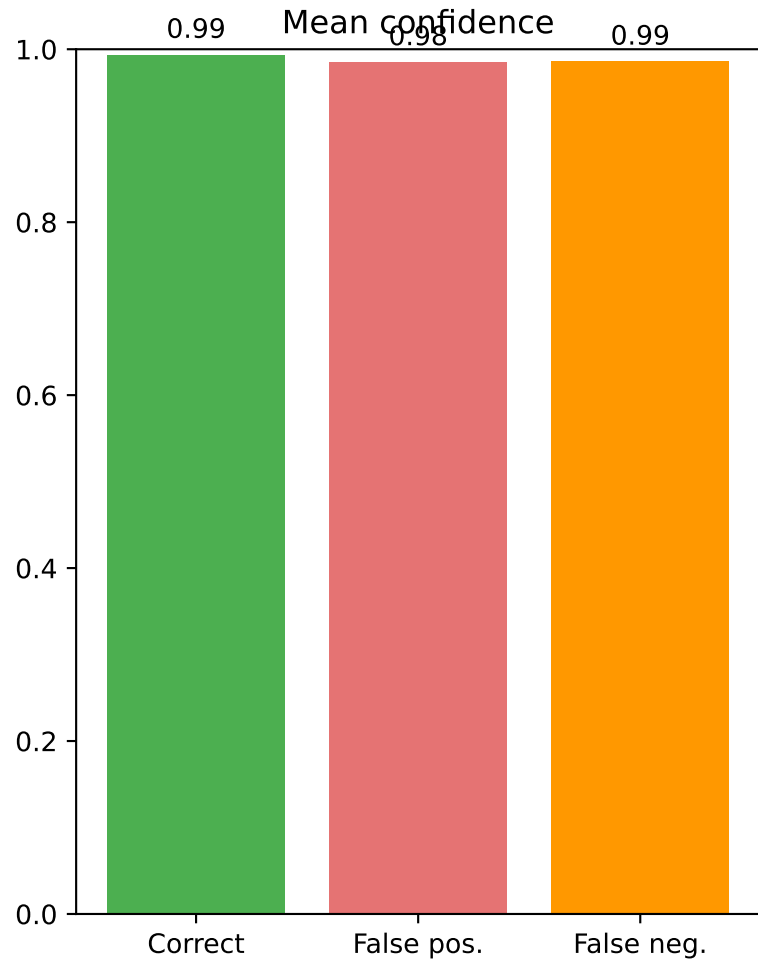
Both cues are over-represented among false positives: reviews that mix a negative clause with a softer/backhanded second clause ("X, but Y") are where the model most often reads the wrong polarity -- the same subtly-negative / sarcastic pattern that motivated Phase 3.

Error analysis -- charts

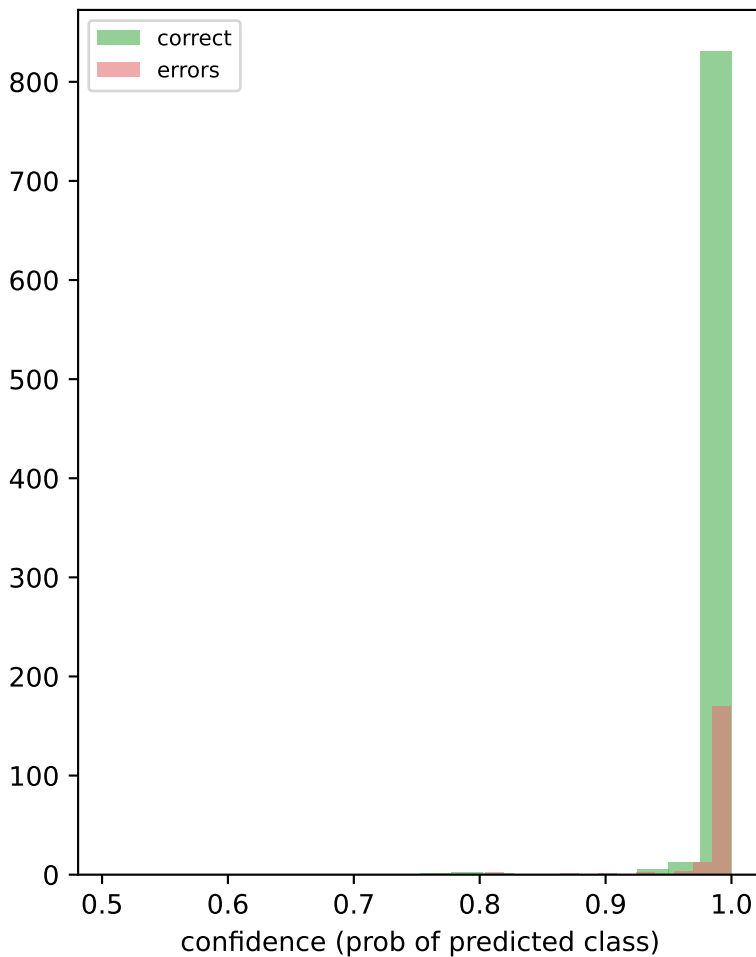
Confusion matrix



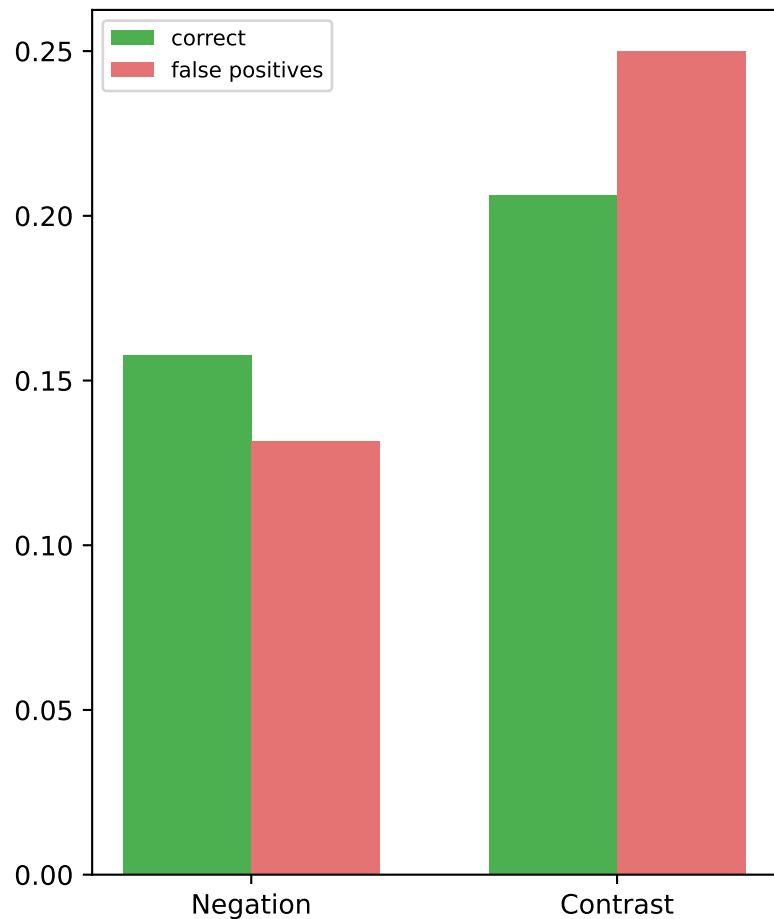
Mean confidence



Confidence distribution



Lexical cue rate



Most confident FALSE POSITIVES (actually negative, predicted positive)

[conf 1.000]
never again , while nothing special , is pleasant , diverting and modest -- definitely a step in the right direction .

[conf 1.000]
been there , done that , liked it much better the first time around - when it was called the professional .

[conf 1.000]
absolutely (and unintentionally) terrifying .

[conf 1.000]
this is a children's film in the truest sense . it's packed with adventure and a worthwhile environmental message , so it's great for the kids . parents , on the other hand , will be ahead of the plot at all times , and there isn't enough clever innuendo to fil

[conf 1.000]
a very familiar tale , one that's been told by countless filmmakers about italian- , chinese- , irish- , latin- , indian- , russian- and other hyphenate american young men struggling to balance conflicting cultural messages .

[conf 1.000]
some motion pictures portray ultimate passion ; others create ultimate thrills . men in black ii achieves ultimate insignificance -- it's the sci-fi comedy spectacle as whiffle-ball epic .

[conf 1.000]
it takes you somewhere you're not likely to have seen before , but beneath the exotic surface (and exotic dancing) it's surprisingly old-fashioned .

[conf 1.000]
harvard man is a semi-throwback , a reminiscence without nostalgia or sentimentality .

[conf 1.000]
the sight of the name bruce willis brings to mind images of a violent battlefield action picture , but the film has a lot more on its mind--maybe too much .

[conf 1.000]
the sweetest thing leaves a bitter taste .

Most confident FALSE NEGATIVES (actually positive, predicted negative)

[conf 0.999]

tells (the story) with such atmospheric ballast that shrugging off the plot's persnickety problems is simply a matter of (being) in a shrugging mood .

[conf 0.999]

maggie smith as the ya-ya member with the o2-tank will absolutely crack you up with her crass , then gasp for gas , verbal deportment .

[conf 0.999]

the movie worked for me right up to the final scene , and then it caved in .

[conf 0.999]

at its worst , the movie is pretty diverting ; the pity is that it rarely achieves its best .

[conf 0.999]

" what really happened ? " is a question for philosophers , not filmmakers ; all the filmmakers need to do is engage an audience .

[conf 0.999]

the main story . . . is compelling enough , but it's difficult to shrug off the annoyance of that chatty fish .

[conf 0.999]

i would be shocked if there was actually one correct interpretation , but that shouldn't make the movie or the discussion any less enjoyable .

[conf 0.999]

devotees of star trek ii : the wrath of khan will feel a nagging sense of deja vu , and the grandeur of the best next generation episodes is lacking .

[conf 0.999]

critics need a good laugh , too , and this too-extreme-for-tv rendition of the notorious mtv show delivers the outrageous , sickening , sidesplitting goods in steaming , visceral heaps .

[conf 0.999]

making such a tragedy the backdrop to a love story risks trivializing it , though chouraqui no doubt intended the film to affirm love's power to help people endure almost unimaginable horror .